

17-9-23

Physics

1) (1) $-(-1)$

2) (2) - on outer surface of the cornea.

3) (4) - virtual, erect, enlarged.

4) (1) - Both A and R are true and R is the correct explanation of A.

5) (1) - Both A and R are true and R is the ~~correct~~ correct explanation ~~of~~ of A.

6) The two most common defects of eye are 1) Myopia 2) Hypermetropia.

1) Myopia can be corrected by using a ~~for~~ concave lens.

2) Hypermetropia can be corrected by using a convex lens.

7) a) A pencil partly immersed in water appears to bent ~~due to reflection~~ ^{refraction} as light rays bent away from

7) a) A pencil partly immersed in water appears to bent $\pm$ due to refraction of light.

b) $n = -2$
 $V = +20 \text{ cm}$
 $f = 9$

$$m = \frac{v}{u}$$

$$-2 = \frac{20}{u}$$

$$u = \frac{-20}{2}$$

$$= -10 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{20} - \frac{1}{-10} = \frac{1}{f}$$

$$\frac{1}{20} + \frac{1}{10} = \frac{1}{f}$$

$$= \frac{1+2}{20} = \frac{1}{f}$$

$$= \frac{3}{20} = \frac{1}{f}$$

$$= \frac{20}{3} = f$$

$$= \underline{\underline{6.67 \text{ cm} = f}}$$

5) $u = -10 \text{ cm}$
focal length = 15 cm

To find :- Image distance V

$$\frac{1}{V} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{V} = \frac{1}{15} + \frac{1}{10} = \frac{25}{150}$$

$$V = 6 \text{ cm}$$

$\therefore$ Image will be formed
at 6 cm beyond the mirror.

$$m = \frac{-V}{u} = \frac{-6}{-10} = 0.6$$

$\therefore$ It will be erect and diminished.

q) M will be negative as the image is real.
Given,
 $u = -10\text{cm}$, $M = -3$, $v = ?$ / $f = ?$

$$M = \frac{v}{u}$$

$$= v = (-3) \times (-10) = 30\text{cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$= \frac{1}{30} - \frac{1}{-10} = \frac{1}{f}$$

$$= \frac{1}{30} + \frac{1}{10} = \frac{1}{f}$$

$$= \frac{1+3}{30} = \frac{4}{30} = \frac{2}{15}$$

$$= f = \pm 7.5\text{cm}$$

- 10) The person is suffering from myopia and hence needs a concave lens to correct the defect. The lens should be such that an object at infinity must form its image at the far point.

$$\text{Hence, } f = -80 \text{ cm} \\ = -0.8 \text{ m}$$

Power of lens can be obtained as:-

$$P = \frac{1}{f}$$

$$P = \frac{1}{(-0.8)} = \boxed{-1.25 \text{ D}}$$

11)

- a) The lens forms a real image on a screen. Therefore, the given lens is a convex lens.
b) Negative as the image is real and inverted.
c) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\frac{1}{20} = \frac{1}{v} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{21} = \frac{(21-20)}{420}$$

$$= \frac{1}{420}$$

$$V = 420 \text{ cm}$$

12)

a) The student should use a convex lens because with the help of a convex lens he will be able to form a real image on the walls.

b) (i) In order to form a magnified image on the wall, the candle flame should be placed between F and $2F$ from the lens.

(ii) In order to form a diminished image on the wall, the candle flame should be placed at a distance greater than $2F$ from the lens.

Q(1)



Q.ii)

